wrong. By closely examining the basic properties of general relativity and quantum mechanics, we can identify what that something is.

The Heart of Quantum Mechanics

When Heisenberg discovered the uncertainty principle, physics turned a sharp corner, never to retrace its steps. Probabilities, wave functions, interference, and quanta all involve radically new ways of seeing reality. Nevertheless, a die-hard "classical" physicist might still have hung on to a thread of hope that when all was said and done these departures would add up to a framework not too distant from old ways of thinking. But the uncertainty principle cleanly and definitively undercut any attempt to cling to the past.

The uncertainty principle tells us that the universe is a frenetic place when examined on smaller and smaller distances and shorter and shorter time scales. We saw some evidence of this in our attempt, described in the preceding chapter, to pinpoint the location of elementary particles such as electrons: By shining light of ever higher frequency on electrons, we measure their position with ever greater precision, but at a cost, since our observations become ever more disruptive. High-frequency photons have a lot of energy and therefore give the electrons a sharp "kick," significantly changing their velocities. Like the frenzy in a room full of children all of whose momentary positions you know with great accuracy but over whose velocities—the speeds and directions in which they are moving—you have almost no control, this inability to know both the positions and velocities of elementary particles implies that the microscopic realm is intrinsically turbulent.

Although this example conveys the basic relationship between uncertainty and frenzy, it actually reveals only part of the story. It might lead you to think, for instance, that uncertainty arises only when we clumsy observers of nature stumble onto the scene. This is not true. The example of an electron violently reacting to being confined in a small box by rattling around at high speed takes us a bit closer to the truth. Even without "direct hits" from an experimenter's disruptive photon, the electron's velocity severely and unpredictably changes from one moment to the next. But even this example does not fully reveal the stunning microscopic features of nature entailed by Heisenberg's discovery. Even in the most quiescent setting imaginable,